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Paper title

When a Restart Corrupts the Result: Duplicate and Missing Records in Dataset Enrichment Pipelines

Thesis

A restart can duplicate or omit records when output and completion are saved separately. I compare replay-safe writes with atomic completion, measuring correctness and recovery cost. The paper is in the six numbered tabs; the remaining tabs provide supporting reference material.

Sections

Section	Core claim
1. The problem	Separate output and completion writes can disagree after a crash
2. Who is affected	Downstream users may receive plausible but incomplete or duplicated data
3. Evidence	Documentation establishes the risk; a synthetic dry run reproduced both failures
4. Existing solutions	Idempotent merge versus atomic progress in the commit
5. Key unknown	Is Delta's transaction id as safe as merge, and cheaper?
6. Individual scope	Four strategies, defined crash points, one laptop
NSF 26-512 summary	What the published opportunity says, section by section
Sources and acronyms	Source addresses and first-use definitions